EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Course Code: MENG245
Course Name: Thermodynamics 1
Semester/ Year: Fall 2023-2024
Instructor: Prof. Ugur Atikol
Assistant: Erfan Malekian
Assessment: Lab 3
Due Date: 2ndjan 2024

LAB NAME: ELECTRIC BOILER

STUDENT(S) NO.: 23704019 **NAME(S), SURNAME(S): TARIK TARIQ**

SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Report Presentation	1, 6	1, 3, 4, 5, 6	10		
Introduction and Procedure	1, 6	1, 3, 4, 5, 6	20		
Results and Calculations	1, 6	1, 3, 4, 5, 6	20		
Discussion & Conclusion	1, 6	1, 3, 4, 5, 6	30		
Answer to Question	1, 6	1, 3, 4, 5, 6	20		
					Marks for report format & referencing are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		**Student Outcomes**						
		1	**2**	**3**	**4**	**5**	**6**	**7**
1		**X**					**X**	
2								
3		**X**					**X**	
4		**X**					**X**	
5		**X**					**X**	
6		**X**					**X**	
	Weight of Student Outcomes	**H**					**H**	

Instructions:

A. **Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.

B. **Submission:** Upload your report as a pdf file to the "Lab Work" section in TEAMS within the designated time. The file uploaded should be named as **[MENG245-Lab3-Student ID-Student Name]**. No late submissions will be accommodated.

ABSTRACT

In this experiment, we studied how water turns into steam in an electric kettle, looking at energy, temperature, and time. We used a smart kettle, measured 1.5 liters of water, timed boiling, and calculated power. We assumed constant pressure, temperatures from 20°C to 100°C, and a 0.1 quality at the end. Enthalpy changes were figured out using tables, giving a power rating of about -3.228 kW. The specific volume at 100°C was 0.1681387 m3/kg. A T-v diagram showed the saturation line and specific volume. While the experiment gave insights into energy and power, some assumptions might affect accuracy. Improvements should focus on better measurements and trying different starting temperatures. The findings offer practical knowledge for system improvements. References include textbooks by Yunus Cengel and Boles for theoretical background, and appendices show T-v diagrams made using tables and Excel.

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LIST OF GRAPHS

1. INTRODUCTION

In this experiment, we're looking at how water turns into steam in a simple electric kettle. We want to understand how much energy it takes and how things like temperature and time play a role. This helps us know more about how things heat up and change form. By measuring temperatures, figuring out energy changes, and keeping an eye on time, we hope to learn useful stuff for making things work better. This report will show how we did the experiment, what we saw, and what it means for everyday things like heating water.

2. APPARATUS USED

The main equipment used is the a water cattle. The water cattle containes a switch control that can be turned on and automacilly switches itself of when the water is completely boiled. it aslo has a smart light indicator that turns on when the cattle is boiling water and turns off when the water is fully boiled. It can accommodate appropriately around 2 litres of water.

Figure 1 electric boiler cattle

3. PRECEDURE

1. Make sure the stainless steel body is place on the power plate for the cattle to work.
2. Measure 1.5 liters of water and pour in the cattle from the mouth.
3. Set your stopwatch and simultaneously switch on the cattle and start the stopwatch.
4. Carefully wait until the smart light indicater turns off and immidiately stop the clock and record the time taken for the water to boil.

4. THEORY

When we heat water, something fascinating happens: it changes from a liquid to a gas. This transformation is special because it soaks up heat energy .What's intriguing is that while water is turning into steam, its temperature doesn't actually go up. It's like it's quietly absorbing all this heat, getting ready to become steam.

assumptions

- We're assuming the pressure remains constant at 1 atmosphere throughout the boiling process.

- At the end of boiling, we're considering the water as a mix of liquid and steam, with about 0.1 to be the quality.

- initial temperature 20 degrees

- final temperature 100 degrees

Power rating of the cattle:

Power = work $_{electric}$/change in time , $P = w/\Delta t$

Using energy equation, Q+W=$m(h_2$-$h_1)$, in this case Q=0 and so our equation reduces to W=$m(h_2$-$h_1)$

To get electric power we divide both sides of the equation by change in time: $\mathrm{W}/\Delta t = \mathrm{m}(\mathrm{h2} - \mathrm{h1})/\Delta t$

Obtaining enthalpies

h_1 can be obtained using the equation: h_1=c_p x T, where the temperature is absolute and c_p can be obtained from the property tables (table A-3, Appendix 1)

h_2(h_{avg}) can be obtained from the property tables using the following equation: h_{avg}=h_f + xh_{fg}, where the quality(x) is given as 0.1.

obtaining the mass of water

there are two ways to calculate mass and both ways will approximately give the same values.

To obtain mass, you can use the following formula, $mass = \frac{volume(V)}{specific\ volume(v)\ at\ initial\ state}$

Or

to obtain the mass of water we use the following formula: $\varrho\ x\ V = m$, where ϱ is the density of water assumed to be 1000kg/m^3 and V is the volume of water (1.5 L).

5. RESULTS AND DISCUSSION

1. Power Rating of the Kettle:

Measured Temperatures:

- Initial temperature : 20°C

- Final temperature : 100°C

Now we are going to use the initial and final temperatures to get the enthalpy values:

$h_1 = c_p$ x T , from the property tables (TABLE-A3) I obtained C_p as 4.188 KJ/kg.K but this value is obtained through interpolating at 20 degrees , now the next step is to convert the temperature from degree celcius to kelvin by adding 273.

$h_1 = c_p$ x T , h_1=4.188 KJ/kg.K x 293K = **1227.084 KJ/kg**

to find h_2, we use the following formula and using h values from the property tables : $\mathbf{h_{avg} = h_f + xh_{fg}}$, where the quality(x) is given as 0.1.

h_2= 419.17KJ/kg + 0.1(2256.4 KJ/kg) = **644.8 KJ/kg**

now, the last thing we need to find before calculating the power rating is the mass. To find the mass we use the following formula: $\boldsymbol{mass = \frac{volume(V)}{specific\ volume(v)\ at\ initial\ state}}$, volume of water that I used was about 1.5L:

$(1.5\cancel{L} \times 1\cancel{m^3}/1000\cancel{L})/0.001002\cancel{m^3}/kg$ = **1.497kg** , note that I converted Liters to meters cubed using the conversion ratio of 1m3=1000L. for the specific volume I obtained it from direct reading on the property tables (TABLE A4) at 20 degrees and used the value of v_f since sometimes compressed liquids may sometimes act like a saturated liquid at low pressure.

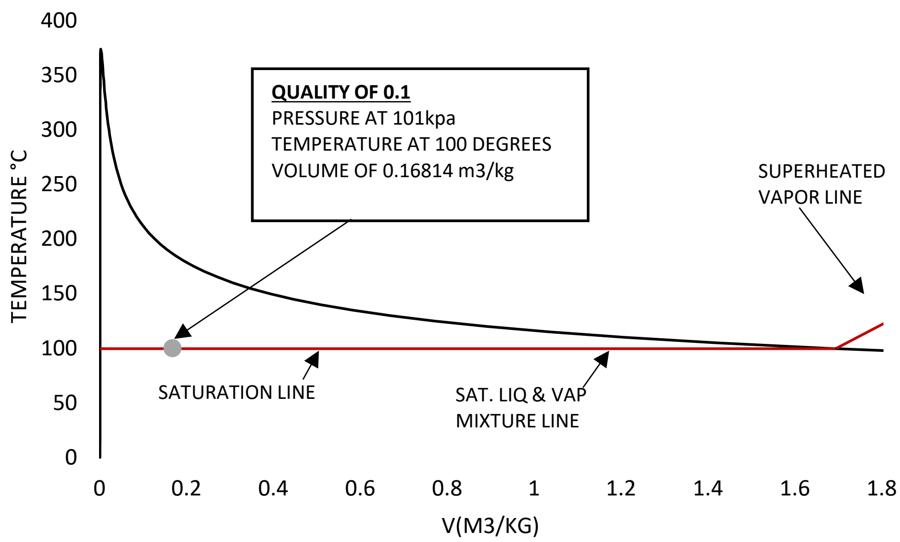
Finally we can now calculated the **power rating** of the cattle using the following formula :

$\boldsymbol{electric\ power} = \frac{\mathbf{m(h2-h1)}}{\Delta t}$ = 1.497kg (644.8KJ/kg – 1227.084KJ/kg)/ 270s = **-3.228KJ/s = -3.228 KW**

Note, I used at stop watch to get the time it will take for the water to boil which was around 270s.

Lastly, I needed to calculate the specific volume as well which I can use to graphically represent it on a T-v diagram shown in figure . To do so I used the formula $v_{final@100°c} = v_f + xv_{fg}$, the v_f and v_{fg} can be obtained from the property table (table A4) . $\mathbf{v_{final@100°c}}$**= 0.001043 m3/kg + 0.1(1.670957m3/kg) = 0.1681387 m3/kg**

Below (graph 1) is a T-v diagram that clearly shows the saturation line at 100 degrees which is the boiling point of water and the specific volume using the o.1 quality.



Graph 1 T-v diagram

DISCUSSION OF RESULTS

In this experiment, our goal was to understand how water turns into steam and figure out the energy needed for this change. We heated water from a starting temperature of 20°C until it boiled, observing and measuring time along the way.

Heat and Energy:

We looked at how much heat energy is involved in this process. Enthalpy is a thermodynamic property of a system. It is often used to quantify the total heat content of a system at constant pressure and is denoted by the symbol H. Enthalpy is particularly useful in the study of processes involving heat transfer, such as chemical reactions and phase changes. Using the property tables, I was able to get the initial and final enthalpies as, $h_1=c_p \times T$, $h_1=$**1227.084KJ/kg** and $\mathbf{h_2=h_f + xh_{fg}}$, where the quality(x) is given as 0.1. $h_2=$ 419.17KJ/kg + 0.1(2256.4 KJ/kg) = **644.8 KJ/kg**

Power:

The power rating, we calculated tells us how fast this change happened. I was able to calculate power using energy equation at constant pressure as **-3.228 KW.** This results were obtained based on certain assumptions and the results obtained may not be a very accurate way to measure the electric power of a cattle to boil water.

Steam Behavior:

We also looked at how much space steam takes up compared to water through calculations of specific volume that I obtained as **0.1681387 m3/kg**. This results can be seen on the P-v diagram for more understanding on how water behaves at different temperatures.

6. CONCLUSION

This experiment helped us learn about the science behind turning water into steam. We found out how much energy is needed, how fast it happens, and how steam behaves.

To sum it up, the steam experiment taught us about water turning into steam and the energy needed for it. However, some guesses we made might affect how accurate our results are. In the future, we can make the experiment better by finding more accurate ways to measure certain things, like how much vapor is in the mix. It would also help if we try the experiment with different starting temperatures to make it more like what happens in real life. Making our measurements more precise and reducing how much heat gets lost during the experiment will give us more accurate results. These improvements not only make our experiment better but also help us understand how this process works in different situations. It's like leveling up our knowledge in how things change from water to steam, which can be super useful in making things work better in real-life situations.

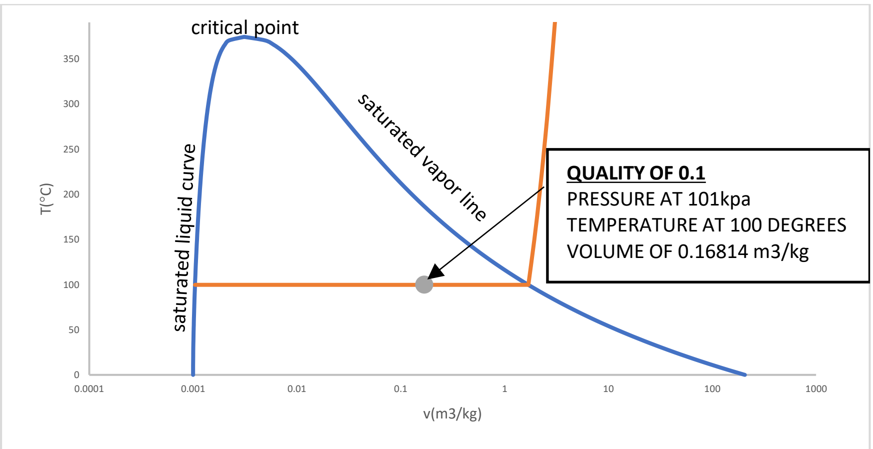
7. REFERENCES

Yunus Cengel and Boles, M. (2012). *Loose Leaf Version for Thermodynamics: An Engineering Approach 7E*. McGraw-Hill Science/Engineering/Math.

Cengel, Y.A. and Boles, M.A. (2015). *Property Tables Booklet for Thermodynamics : an Engineering Approach*. Boston: Mcgraw Hill.

8. APPENDIX

The graph below shows another T-v diagram that i constructed on excel by using the property tables available in our books. This was graph shows the saturated liquid curve up to the critical point as well as the saturated vapor line. I managed to demostrate these points by using the logarithmic scale available on excel where the x axis is compressed. The saturation line is also clearly shown and was constructed by making good use of superheated values from the property tables.



Graph 2 T-v diagram with compressed x-axis